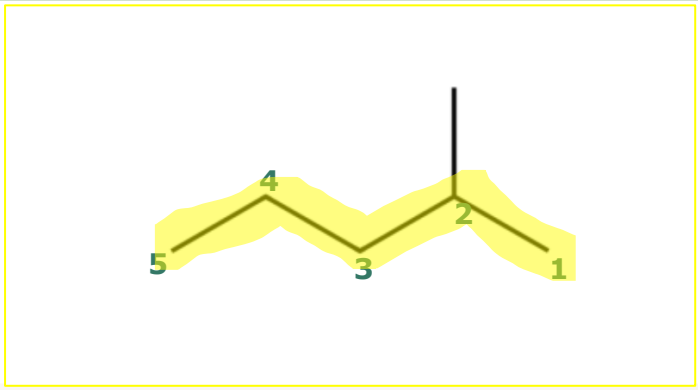
A01 STRUCTURE → NAME

PART A · Counted molecular formula: C_6H_{14}

01 NUMBERED PARENT



02 NAMING TRAIL

Parent: 5 C → pentane; open chain.

Numbering: Right → left (2); reverse (4).

Choice: 2 < 4, so start at the right-hand end.

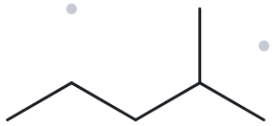
Substituents: C2 methyl → 2-methyl.

03 MY NAME · IUPAC

2-methylpentane

5 parent C + 1 methyl C = 6 C; 2(6)+2 = 14 H.

04 THE ALKANE CHECK



2-methylpentane

Returned: 2-methylpentane

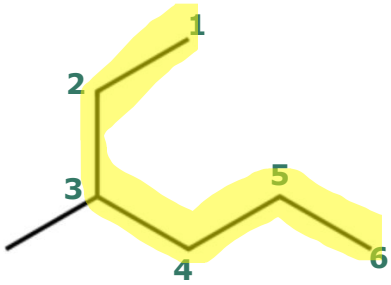
MATCH — structure, name and formula agree.

✓ Parent confirmed ✓ Lowest locants ✓ Alphabetical order ✓ Punctuation ✓ Formula counted

A02 STRUCTURE → NAME

PART A · Counted molecular formula: C_7H_{16}

01 NUMBERED PARENT



02 NAMING TRAIL

Parent: 6 C → hexane; the longest path bends upward.

Numbering: Upper end → right (3); reverse(4).

Choice: 3 < 4; include the upper two-carbon arm.

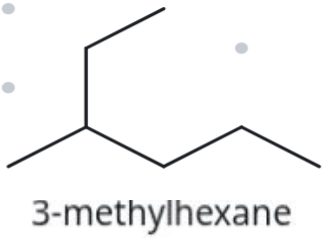
Substituents: C3 methyl → 3-methyl.

03 MY NAME · IUPAC

3-methylhexane

6 parent C + 1 methyl C = 7 C; 2(7)+2 = 16 H.

04 THE ALKANE CHECK



Returned: 3-methylhexane

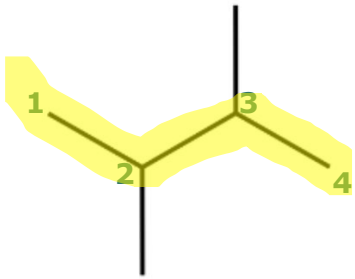
MATCH — structure, name and formula agree.

✓ Parent confirmed ✓ Lowest locants ✓ Alphabetical order ✓ Punctuation ✓ Formula counted

A03 STRUCTURE → NAME

PART A · Counted molecular formula: C_6H_{14}

01 NUMBERED PARENT



02 NAMING TRAIL

Parent: 4 C → butane; open chain.

Numbering: Left → right (2,3)reverse 2,3).

Choice: Both directions give the same lowest locants.

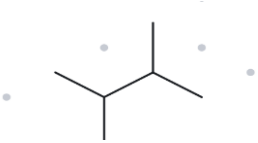
Substituents: C2 and C3 methyl → 2,3-dimethyl.

03 MY NAME · IUPAC

2,3-dimethylbutane

4 parent C + 2 methyl C = 6 C; 2(6)+2 = 14 H.

04 THE ALKANE CHECK



2,3-dimethylbutane

Returned: 2,3-dimethylbutane

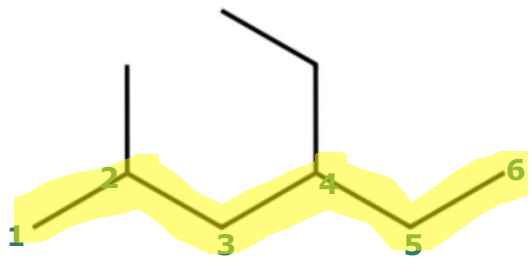
MATCH — structure, name and formula agree.

✓ Parent confirmed ✓ Lowest locants ✓ Alphabetical order ✓ Punctuation ✓ Formula counted

A04 STRUCTURE → NAME

PART A · Counted molecular formula: C_9H_{20}

01 NUMBERED PARENT



02 NAMING TRAIL

Parent: 6 C → hexane; choose a longest chain.

Numbering: Left → right (2,4); reverse (3,5).

Choice: At the first difference, 2 < 3.

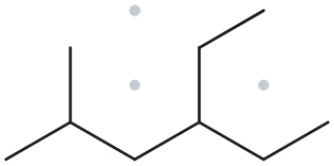
Substituents: C4 ethyl; C2 methyl (ethyl before methyl).

03 MY NAME · IUPAC

4-ethyl-2-methylhexane

6 + 2 (ethyl) + 1 (methyl) = 9 C; 2(9)+2 = 20 H.

04 THE ALKANE CHECK



4-ethyl-2-methylhexane

Returned: 4-ethyl-2-methylhexane

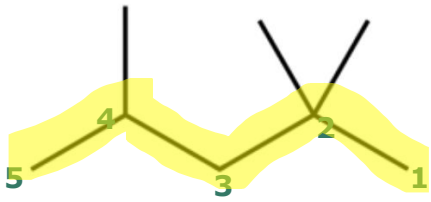
MATCH — structure, name and formula agree.

✓ Parent confirmed ✓ Lowest locants ✓ Alphabetical order ✓ Punctuation ✓ Formula counted

A05 STRUCTURE → NAME

PART A · Counted molecular formula: C_8H_{18}

01 NUMBERED PARENT



02 NAMING TRAIL

Parent: 5 C → pentane; open chain.

Numbering: Right → left {2,2,4}; reverse {2,4,4}.

Choice: First locants tie; the second 2 beats 4.

Substituents: C2, C2 and C4 methyl → 2,2,4-trimethyl.

03 MY NAME · IUPAC

2,2,4-trimethylpentane

5 parent C + 3 methyl C = 8 C; 2(8)+2 = 18 H.

04 THE ALKANE CHECK



2,2,4-trimethylpentane

Returned: 2,2,4-trimethylpentane

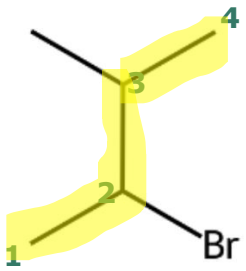
MATCH — structure, name and formula agree.

✓ Parent confirmed ✓ Lowest locants ✓ Alphabetical order ✓ Punctuation ✓ Formula counted

A06 STRUCTURE → NAME

PART A · Counted molecular formula: $C_5H_{11}Br$

01 NUMBERED PARENT



02 NAMING TRAIL

Parent: 4 C → butane; open chain.

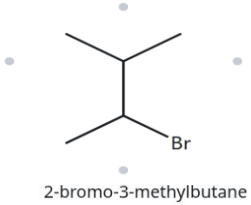
Numbering: Lower end: Br 2, methyl 3; reverse: Br 3, methyl 2.
Choice: {2,3} ties; bromo precedes methyl, so Br gets 2.
Substituents: C2 bromo; C3 methyl (b before m).

03 MY NAME · IUPAC

2-bromo-3-methylbutane

5 C gives C_5H_{12}; replace one H with Br → $C_5H_{11}Br$.

04 THE ALKANE CHECK



2-bromo-3-methylbutane

Returned: 2-bromo-3-methylbutane

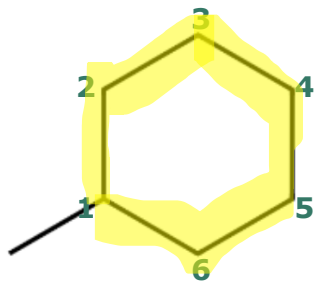
MATCH — structure, name and formula agree.

✓ Parent confirmed ✓ Lowest locants ✓ Alphabetical order ✓ Punctuation ✓ Formula counted

A07 STRUCTURE → NAME

PART A · Counted molecular formula: C_7H_{14}

01 NUMBERED PARENT



02 NAMING TRAIL

Parent: 6 C ring → cyclohexane.

Numbering: Start at the substituted carbon: 1 either way.
Choice: One substituent needs no locant in the final name.

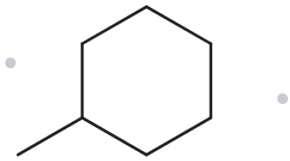
Substituents: One methyl group on the ring.

03 MY NAME · IUPAC

methylcyclohexane

6 ring C + 1 methyl C = 7 C; one ring → 2(7) = 14 H.

04 THE ALKANE CHECK



methylcyclohexane

Returned: methylcyclohexane

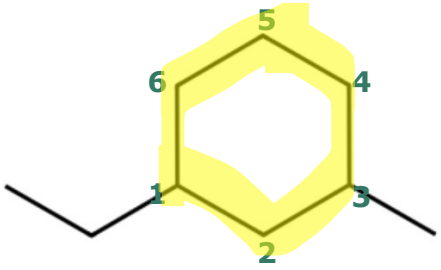
MATCH — structure, name and formula agree.

✓ Parent confirmed ✓ Lowest locants ✓ Alphabetical order ✓ Punctuation ✓ Formula counted

A08 STRUCTURE → NAME

PART A · Counted molecular formula: C_9H_{18}

01 NUMBERED PARENT



02 NAMING TRAIL

Parent: 6 C ring → cyclohexane.

Numbering: From ethyl: (1,3) vs (1,5); from methyl: (1,3).

Choice: Choose 1,3; ethyl gets 1 in the alphabetical tie.

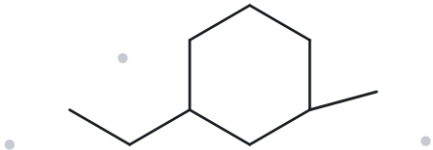
Substituents: C1 ethyl; C3 methyl (ethyl before methyl).

03 MY NAME · IUPAC

1-ethyl-3-methylcyclohexane

6 + 2 (ethyl) + 1 (methyl) = 9 C; one ring → 18 H.

04 THE ALKANE CHECK



Returned: 1-ethyl-3-methylcyclohexane

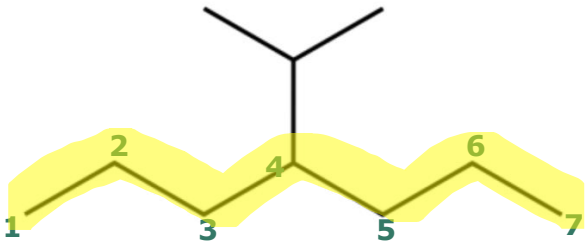
MATCH — structure, name and formula agree.

✓ Parent confirmed ✓ Lowest locants ✓ Alphabetical order ✓ Punctuation ✓ Formula counted

A09 STRUCTURE → NAME

PART A · Counted molecular formula: $C_{10}H_{22}$

01 NUMBERED PARENT



02 NAMING TRAIL

Parent: 7 C → heptane; trace the entire lower zigzag.

Numbering: Left → right {4}; reverse {4}.

Choice: Both directions put the branched group at C4.
Substituents: C4 propan-2-yl, $CH(CH_3)2$; also called isopropyl.

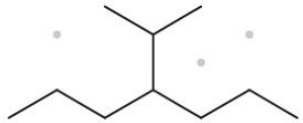
03 MY NAME · IUPAC

4-(propan-2-yl)heptane

7 parent C + 3 substituent C = 10 C; 2(10)+2 = 22 H.

.

04 THE ALKANE CHECK



2-methyl-3-propylhexane

Returned: 2-methyl-3-propylhexane

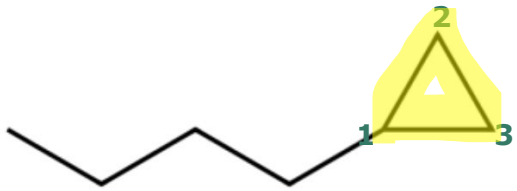
MISMATCH

✓ Parent confirmed ✓ Lowest locants ✓ Alphabetical order ✓ Punctuation ✓ Formula counted

A10 STRUCTURE → NAME

PART A · Counted molecular formula: C_7H_{14}

01 NUMBERED PARENT



02 NAMING TRAIL

Parent: 3 C ring → cyclopropane (activity ring-parent rule).
Numbering: Attachment carbon is 1, clockwise or anticlockwise.
Choice: The ring is the parent; a single locant is omitted.

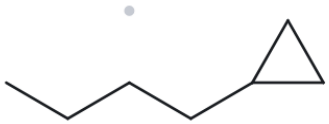
Substituents: One unbranched 4-C butyl group.

03 MY NAME · IUPAC

butylcyclopropane

3 ring C + 4 butyl C = 7 C; one ring → 2(7) = 14 H.

04 THE ALKANE CHECK



butylcyclopropane

Returned: butylcyclopropane

MATCH — structure, name and formula agree.

✓ Parent confirmed ✓ Lowest locants ✓ Alphabetical order ✓ Punctuation ✓ Formula counted

B01 NAME → STRUCTURE

PART B · GIVEN NAME: 2-methylpropane

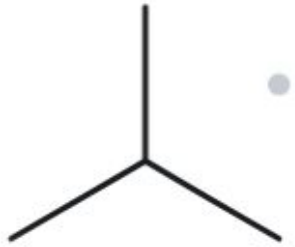
01 NAME DECODED

Parent: prop- → 3-carbon chain: propane.

Substituents: One methyl group at C2.

Expected formula: 3 + 1 = 4 C; open chain → C_4H_{10}.

02 BOND-LINE STRUCTURE



03 FORMULA CHECK

Counted: 3 CH_3 + 1 CH = C_4H_{10}

C_nH_{2n+2}: 2(4)+2 = 10 H.

Formula agrees: C_4H_{10} ✓

04 BACK-CHECK

THE ALKANE RETURNED

2-methylpropane

MATCH

The longest path has 3 C; methyl is at C2.

✓ Parent checked ✓ Positions checked ✓ Formula agrees ✓ Back-check recorded

B02 NAME → STRUCTURE

PART B · GIVEN NAME: 3-ethylpentane

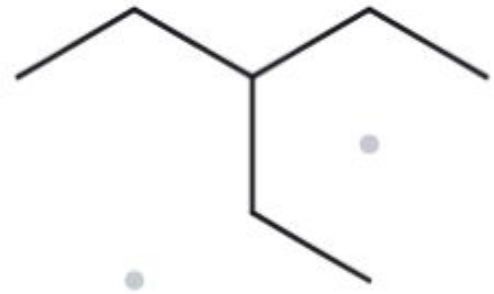
01 NAME DECODED

Parent: pent- → 5-carbon chain: pentane.

Substituents: One ethyl group (2 C) at C3.

Expected formula: 5 + 2 = 7 C; open chain → C_7H_{16}.

02 BOND-LINE STRUCTURE



03 FORMULA CHECK

Counted: 3 CH_3 + 3 CH_2 + 1 CH = C_7H_{16}.

C_nH_{2n+2}: 2(7)+2 = 16 H.

Formula agrees: C_7H_{16} ✓

04 BACK-CHECK

THE ALKANE RETURNED

3-ethylpentane

MATCH

All three arms are 2 C long; the parent has 5 C.

✓ Parent checked ✓ Positions checked ✓ Formula agrees ✓ Back-check recorded

B03 NAME → STRUCTURE

PART B · GIVEN NAME: 2,2-dimethylbutane

01 NAME DECODED

Parent: but- → 4-carbon chain: butane.

Substituents: Two methyl groups, both on C2.

Expected formula: 4 + 1 + 1 = 6 C → C_6H_{14}.

02 BOND-LINE STRUCTURE



03 FORMULA CHECK

Counted: 4 CH_3 + 1 CH_2 + 1 C = C_6H_{14}

C_nH_{2n+2}: 2(6)+2 = 14 H.

Formula agrees: C_6H_{14} ✓

04 BACK-CHECK

THE ALKANE RETURNED

2,2-dimethylbutane

MATCH

C2 has four C–C bonds and no hydrogen.

✓ Parent checked ✓ Positions checked ✓ Formula agrees ✓ Back-check recorded

B04 NAME → STRUCTURE

PART B · GIVEN NAME: 3-ethyl-2-methylpentane

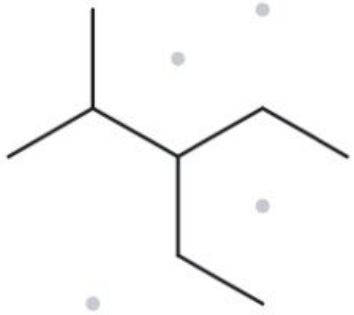
01 NAME DECODED

Parent: pent- → 5-carbon chain: pentane.

Substituents: C3 ethyl (2 C); C2 methyl (1 C).

Expected formula: 5 + 2 + 1 = 8 C → C_8H_{18}.

02 BOND-LINE STRUCTURE



03 FORMULA CHECK

Counted: 4 CH_3 + 2 CH_2 + 2 CH = C_8H_{18}.

C_nH_{2n+2}: 2(8)+2 = 18 H.

Formula agrees: C_8H_{18} ✓

04 BACK-CHECK

THE ALKANE RETURNED

3-ethyl-2-methylpentane

MATCH

Choose a 5-C chain with two substituents; locants 2,3.

✓ Parent checked ✓ Positions checked ✓ Formula agrees ✓ Back-check recorded

B05 NAME → STRUCTURE

PART B · GIVEN NAME: 1-chloro-2-methylpropane

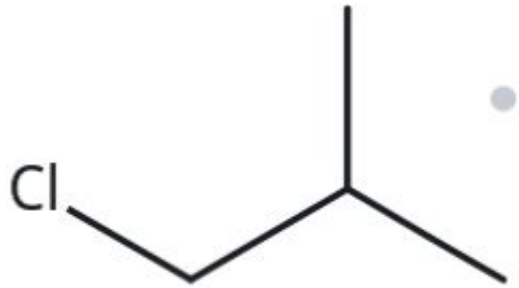
01 NAME DECODED

Parent: prop- → 3-carbon chain: propane.

Substituents: C1 chloro; C2 methyl.

Expected formula: 4 C → C_4H_{10}; replace 1 H with Cl → C_4H_9Cl.

02 BOND-LINE STRUCTURE



03 FORMULA CHECK

Counted: 2 CH_3 + CH + CH_2Cl = C_4H_9Cl.

Open chain: 2(4)+2−1 = 9 H; one Cl.

Formula agrees: C_4H_9Cl ✓

04 BACK-CHECK

THE ALKANE RETURNED

1-chloro-2-methylpropane

MATCH

Cl is on a terminal carbon; chloro precedes methyl.

✓ Parent checked ✓ Positions checked ✓ Formula agrees ✓ Back-check recorded

B06 NAME → STRUCTURE

PART B · GIVEN NAME: 1,2-dimethylcyclopentane

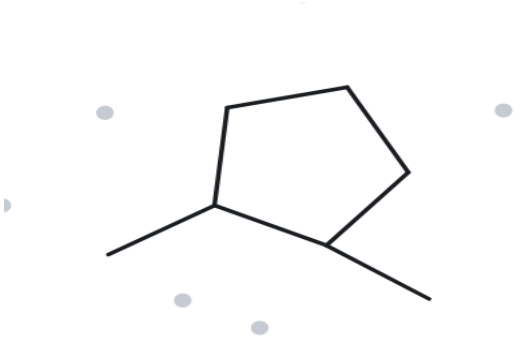
01 NAME DECODED

Parent: cyclopent– → a 5-carbon ring.

Substituents: Methyl groups at adjacent ring carbons C1 and C2.

Expected formula: 5 + 2 = 7 C; one ring → C_7H_{14}.

02 BOND-LINE STRUCTURE



03 FORMULA CHECK

Counted: 2 CH_3 + 3 CH_2 + 2 CH = C_7H_{14}.

C_nH_{2n}: 2(7) = 14 H.

Formula agrees: C_7H_{14} ✓

04 BACK-CHECK

THE ALKANE RETURNED

1,2-dimethylcyclopentane

MATCH

Use 1,2 rather than 1,5; stereochemistry is not required.

✓ Parent checked ✓ Positions checked ✓ Formula agrees ✓ Back-check recorded

B07 NAME → STRUCTURE

PART B · GIVEN NAME: 4-ethyl-3,3-dimethylheptane

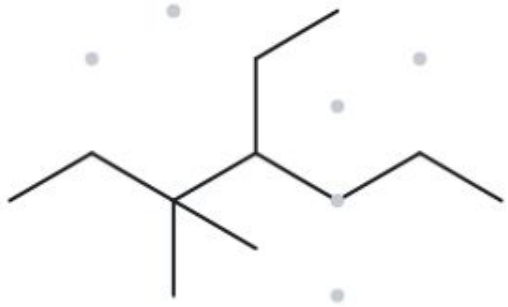
01 NAME DECODED

Parent: hept- → 7-carbon chain: heptane.

Substituents: C4 ethyl; two methyl groups at C3.

Expected formula: 7 + 2 + 1 + 1 = 11 C → $C_{11}H_{24}$.

02 BOND-LINE STRUCTURE



03 FORMULA CHECK

Counted: 5 CH_3 + 4 CH_2 + 1 CH + 1 C = $C_{11}H_{24}$.

C_nH_{2n+2}: 2(11)+2 = 24 H.

Formula agrees: $C_{11}H_{24}$ ✓

04 BACK-CHECK

THE ALKANE RETURNED

4-ethyl-3,3-dimethylheptane

MATCH

Locants 3,3,4 beat 4,5,5; ethyl is cited before methyl.

✓ Parent checked ✓ Positions checked ✓ Formula agrees ✓ Back-check recorded

B08 NAME → STRUCTURE

PART B · GIVEN NAME: tert-butylcyclohexane

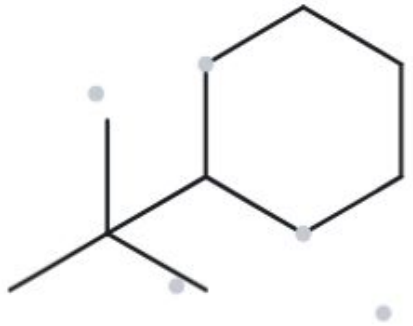
01 NAME DECODED

Parent: cyclohex- → a 6-carbon ring.

Substituents: One tert-butyl group: ring-$C(CH_3)3$.

Expected formula: 6 + 4 = 10 C; one ring → $C_{10}H_{20}$.

02 BOND-LINE STRUCTURE



03 FORMULA CHECK

Counted: 3 CH_3 + 5 CH_2 + 1 CH + 1 C = $C_{10}H_{20}$.

C_nH_{2n}: 2(10) = 20 H.

Formula agrees: $C_{10}H_{20}$ ✓

04 BACK-CHECK

THE ALKANE RETURNED

(1,1-dimethylethyl)cyclohexane

MATCH · ACCEPTED EQUIVALENT

EQUIVALENT – tert-butyl and (1,1-dimethylethyl) both mean $-C(CH_3)3$. Connectivity agrees.

✓ Parent checked ✓ Positions checked ✓ Formula agrees ✓ Back-check recorded

B09 NAME → STRUCTURE

PART B · GIVEN NAME: 3-ethyl-1,1-dimethylcyclohexane

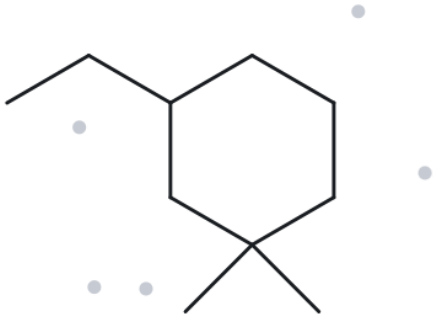
01 NAME DECODED

Parent: cyclohex- → a 6-carbon ring.

Substituents: C3 ethyl; two methyl groups on ring carbon C1.

Expected formula: 6 + 2 + 1 + 1 = 10 C → $C_{10}H_{20}$.

02 BOND-LINE STRUCTURE



03 FORMULA CHECK

Counted: 3 CH_3 + 5 CH_2 + 1 CH + 1 C = $C_{10}H_{20}$.

C_nH_{2n}: 2(10) = 20 H.

Formula agrees: $C_{10}H_{20}$ ✓

04 BACK-CHECK

THE ALKANE RETURNED

3-ethyl-1,1-dimethylcyclohexane

MATCH

1,1,3 beats 1,3,3 and 1,1,5. Cite ethyl first.

✓ Parent checked ✓ Positions checked ✓ Formula agrees ✓ Back-check recorded

B10 NAME → STRUCTURE

PART B · GIVEN NAME: 2-ethylbutane (deliberately incorrect)

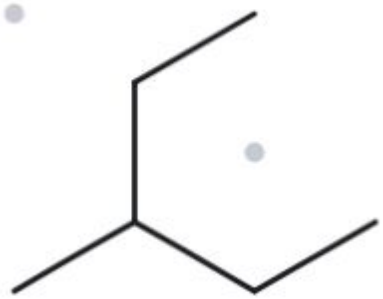
01 NAME DECODED

Parent: As written: but– → 4–carbon chain (butane).

Substituents: As written: one ethyl group at C2.

Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}.

02 BOND-LINE STRUCTURE



03 FORMULA CHECK

Counted: 3 CH_3 + 2 CH_2 + 1 CH = C_6H_{14}.

C_nH_{2n+2}: 2(6)+2 = 14 H.

Formula agrees: C_6H_{14} ✓

04 BACK-CHECK

THE ALKANE RETURNED

3-methylpentane

MISMATCH · GIVEN NAME CORRECTED

Correct name: 3–methylpentane; methyl is at C3.

✓ Parent checked ✓ Positions checked ✓ Formula agrees ✓ Back-check recorded